UPCAT REVIEWER

SOLUTION TO

Practice Test 1

UPCAT REVIEWER

PRACTICE TEST 1

SOLUTION

1. Find the contrapositive of the following statement. “If a figure has three sides, it is a triangle.”

(A) If a figure does not have three sides, it is a triangle.
(B) If a figure is a triangle, then it does not have three sides
(C) If a figure is not a triangle, then it does not have three sides.
(D) If a figure has three sides, it is not a triangle.

Solution:

Recall:

The contrapositive of the statement “If p then q” is “If not q then not p”

So the contrapositive of “If a figure has three sides, it is a triangle” is
“If a figure is not a triangle, then it does not have three sides”

Answer: C

2. Solve for x: $\sqrt{x+6}+\sqrt{x}=4$

(A) no solution (B) 100 (C) 5 (D) $\frac{25}{16}$

Solution:

$\sqrt{x+6}+\sqrt{x}=4$ transpose $\sqrt{x}$ to the right

$\sqrt{x+6}=4-\sqrt{x}$ get the square of both sides

SOLUTION

$\cancel{x}+6=16-8\sqrt{x}+\cancel{x}$

$8\sqrt{x}=10$ divide both sides by 8

$\sqrt{x}=\frac{5}{4}$ get the square of both sides

$x=\frac{25}{16}$

Checking:

$\sqrt{x+6}+\sqrt{x}=4$

$\sqrt{\frac{25}{16}+6}+\sqrt{\frac{25}{16}}\stackrel{?}{=}4$

$\sqrt{\frac{25+96}{16}}+\sqrt{\frac{25}{16}}\stackrel{?}{=}4$

$\sqrt{\frac{121}{16}}+\sqrt{\frac{25}{16}}\stackrel{?}{=}4$

$\frac{11}{4}+\frac{5}{4}\stackrel{?}{=}4$

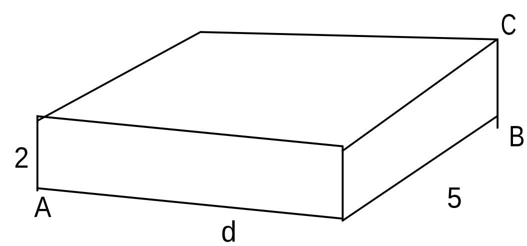
$\frac{16}{4}\stackrel{?}{=}4$

$4\stackrel{\checkmark}{=}4$

Answer: D

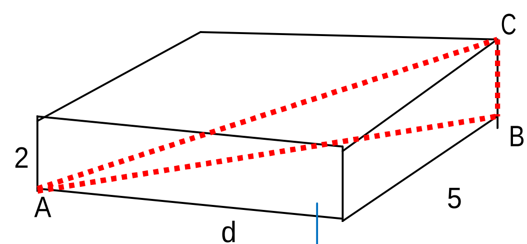
SOLUTION

3. Find the length of diagonal $\overline{AC}$ in the rectangular solid shown. Dimensions are in feet.



(A) $29+d^2$ ft (B) $7+d$ ft (C) $\sqrt{29+d^2}$ ft (D) $\sqrt{7+d}$ ft

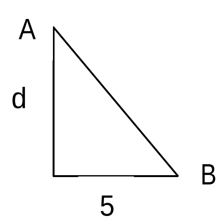
Solution:



Using Pythagorean Theorem

$$d^2+5^2=(AB)^2$$

$$\Rightarrow AB=\sqrt{d^2+25}$$



Now, we apply the Pythagorean Theorem to ΔABC

$$(AB)^2+(BC)^2=(AC)^2$$

SOLUTION

$$\left(d^2+25\right)+2^2=\left(AC\right)^2$$

$$\left(AC\right)^2=d^2+29$$

$$AC=\sqrt{d^2+29}$$

Answer: C

4. The area of a regular octagon is $30\,cm^2$. What is the area of a regular octagon with sides four times as large?

(A) $545\ cm^2$ (B) $480\ cm^2$ (C) $3600\ cm^2$ (D) $120\ cm^2$

Solution:

Given: $A_1=30\ cm^2$

$$A_2=?$$

$s_2=4s_1$ or $\dfrac{s_2}{s_1}=4$

$$\frac{A_2}{A_1}=\left(\frac{s_2}{s_1}\right)^2$$

$$\frac{A_2}{A_1}=4^2$$

$$A_2=16\,A_1$$

$$A_2=16\left(30\right)$$

$$A_2=480\ cm^2$$

Answer: B

SOLUTION

5. Simplify: $\left(\sqrt{3}-\sqrt{7}\right)\left(\sqrt{3}+\sqrt{7}\right)$

(A) -4 (B) 58 (C) 10 (D) -40

Solution:

$$\left(\sqrt{3}-\sqrt{7}\right)\left(\sqrt{3}+\sqrt{7}\right)=\left(\sqrt{3}\right)^2-\left(\sqrt{7}\right)^2$$
$$=3-7$$
$$=-4$$

Recall:

$$(a-b)(a+b)=a^2-b^2$$

Answer: A

6. If the sum of the roots of $x^2+3x-5=0$ is added to the product of its roots, the result is

(A) -2 (B) -8 (C) -15 (D) 15

SOLUTION

Solution:

Recall:

In the quadratic equation

$ax^2 + bx + c = 0$, where $a \neq 0$

Sum of roots = $-\dfrac{b}{a}$

Product of roots = $\dfrac{c}{a}$

Why?

The roots of $ax^2 + bx + c = 0$ using quadratic formula are:

$$\frac{-b-\sqrt{b^2-4ac}}{2a} \quad \text{and} \quad \frac{-b+\sqrt{b^2-4ac}}{2a}$$

Sum of roots = $\dfrac{-b-\sqrt{b^2-4ac}}{2a} + \dfrac{-b+\sqrt{b^2-4ac}}{2a}$

$= \dfrac{-2b}{2a}$

$= \dfrac{-b}{a}$

product of roots =

$$\left(\frac{-b-\sqrt{b^2-4ac}}{2a}\right)\left(\frac{-b+\sqrt{b^2-4ac}}{2a}\right)$$

$$= \frac{(-b)^2-\left(\sqrt{b^2-4ac}\right)^2}{(2a)^2}$$

$$= \frac{b^2-\left(b^2-4ac\right)}{(2a)^2}$$

$$= \frac{4ac}{4a^2}$$

$$= \frac{c}{a}$$

In $x^2 + 3x - 5 = 0$ a=1, b=3, and c=-5

So

Sum of roots = $-\dfrac{3}{1}$ = -3

Product of roots = $\dfrac{-5}{1}$ = -5

Sum + Product of roots = (-3) + (-5) = -8

Answer: B

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7. The roots of the equation $2x^2 - x = 4$ are

   (A) real, rational, and unequal
   (B) real and irrational
   (C) real, rational, and equal
   (D) imaginary

Solution:

Express $2x^2 - x = 4$ in the form $ax^2 + bx + c = 0$, and compute the discriminant $b^2 - 4ac$

The $b^2 - 4ac$ term is called the discriminant and it helps to determine how many and what kind of roots you see in the solution

Value of $b^2 - 4ac$	Is it a perfect square?	Nature of the Roots
b2 – 4ac > 0	yes	2 real roots, rational
b2 – 4ac > 0	no	2 real roots, irrational
b2 – 4ac < 0	not possible	2 imaginary roots
b2 – 4ac = 0	not possible	1 real root

So $2x^2 - x = 4$ becomes $2x^2 - x - 4 = 0$, a = 2, b = -1, c = -4

$$b^2 - 4ac = (-1)^2 - 4(2)(-4) = 33$$

The result is 33 which is greater than zero and not a perfect square.

therefore the roots are r**eal and irrational.**

Answer: B

SOLUTION

8. Which statement must be true if a parabola represented by the equation $y = ax^2 + bx + c$ does not intersect the x-axis?

(A) $b^2 - 4ac > 0,$ and $b^2 - 4ac$ is not a perfect square

(B) $b^2 - 4ac > 0,$ and $b^2 - 4ac$ is a perfect square

(C) $b^2 - 4ac < 0$

(D) $b^2 - 4ac = 0$

Solution:

To get the x-intercept/s of $y = ax^2 + bx + c$ we let y = 0, and solve for x.

$y = ax^2 + bx + c$ does not intersect the x-axis when the roots of $ax^2 + bx + c = 0$ are not real or imaginary.

That happens when $b^2 - 4ac < 0$

Answer: C

SOLUTION

9. The value of $\left(\dfrac{3^0}{27^{\frac{2}{3}}}\right)^{-1}$ is

(A) -9 (B) $-\dfrac{1}{9}$ (C) 9 (D) $\dfrac{1}{9}$

Solution:

$$\left(\frac{3^0}{27^{\frac{2}{3}}}\right)^{-1} = \left(\frac{1}{27^{\frac{2}{3}}}\right)^{-1}$$

$$= 27^{\frac{2}{3}}$$

$$= \left(\sqrt[3]{27}\right)^2$$

$$= 3^2$$

$$= 9$$

Answer: C

10. What is the last term in the expansion of $(x+2y)^5$?

(A) $2y^5$ (B) $32y^5$ (C) y^5 (D) $10y^5$

Solution:

The last term in the expansion of $(x+2y)^5$ is $(2y)^5$

which is equal to $32y^5$

Answer: B

SOLUTION

11. The larger root of the equation $(x+3)(x-4)=0$ is

(A) -3 (B) -4 (C) 4 (D) 3

Solution:

$(x+3)(x-4)=0$

$x+3=0$ or $x-4=0$

$x=-3$ or $x=4$

The larger of -3 and 4 is 4.

Answer: C

12. Express $\frac{1}{x+1}+\frac{1}{x}$ as a single fraction.

(A) $\frac{2x+3}{x^2+x}$ (B) $\frac{2x+1}{x^2+x}$ (C) $\frac{2}{2x+1}$ (D) $\frac{3}{x^2}$

Solution:

$$\frac{1}{x+1}+\frac{1}{x}=\frac{x+(x+1)}{x(x+1)}$$

$$=\frac{2x+1}{x(x+1)}$$

$$=\frac{2x+1}{x^2+x}$$

Answer: B

SOLUTION

13. Ano ang kabuuan ng walang katapusang geometric series na
$3.1 + 1.86 + 1.116 + 0.6696 + \ldots$?

(A) 8.75 (B) 9.75 (C) 4.75 (D) 7.75

Solution:

Let $x = 3.1 + 1.86 + 1.116 + 0.6696 + \ldots$

$$x = 3.1 + 0.6\left(3.1 + 1.86 + 1.116 + \ldots\right)$$

$$x = 3.1 + 0.6\,x$$

$$0.4x = 3.1$$

$$x = \frac{3.1}{0.4}$$

$$x = \frac{31}{4} \text{ or } 7.75$$

We can also use the formula $S_n = \dfrac{a_1}{1-r}$, where a_1 = first term and r = common ratio

$$S_n = \frac{3.1}{1-0.6}$$

$$S_n = \frac{3.1}{0.4} = \frac{31}{4} = 7.75$$

Answer: D

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14. Which equation represents a hyperbola?

(A) $y = 16x^2$ (B) $y = 16 - x^2$ (C) $y^2 = 16 - x^2$ (D) $y = \dfrac{16}{x}$

Solution:

(A) $y = 16x^2$ is a parabola

> Recall:
>
> The graph of the quadratic function $y = ax^2 + bx + c$, where a, b, and c are constants and $a \neq 0$, is a parabola that opens upward if a>0, and a parabola that opens downward if a<0.

(B) $y = 16 - x^2$ is also a parabola and it opens downward.

(C) $y^2 = 16 - x^2$ can be expressed as $x^2 + y^2 = 16$.

Therefore it is a circle.

> Recall:
>
> The graph of the equation $(x-h)^2 + (y-k)^2 = r^2$ is a circle whose radius is r and center is (h,k)

(D) $y = \dfrac{16}{x}$ Its graph is a hyperbola.

> Recall:
>
> If y varies inversely as x, then $y = \dfrac{k}{x}$, where k is a constant of variation and its graph is a hyperbola. (in $y = \dfrac{16}{x}$,k=16)

Answer: D

SOLUTION

15. Which expression is equivalent to the complex fraction $\dfrac{\dfrac{x}{x+2}}{1-\dfrac{x}{x+2}}$?

(A) $\dfrac{x}{2}$ (B) $\dfrac{2x}{x+2}$ (C) $\dfrac{2x}{x^2+4}$ (D) $\dfrac{2}{x}$

Solution:

$$\frac{\left(\dfrac{x}{x+2}\right)}{\left(1-\dfrac{x}{x+2}\right)} \bullet \frac{x+2}{x+2} = \frac{x}{(x+2)-x}$$

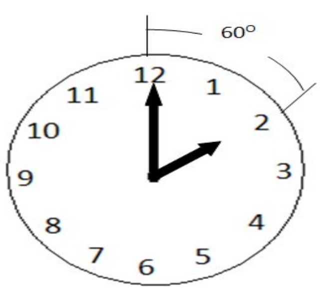
$$= \frac{x}{2}$$

Answer: A

16. What is the radian measure of the angle formed by the hands of the clock at 2:00 pm?

(A) $\dfrac{\pi}{2}$ (B) $\dfrac{\pi}{3}$ (C) $\dfrac{\pi}{4}$ (D) $\dfrac{\pi}{6}$

Solution:



The degree measure formed by the hands of the clock at 2:00 is 60^O. To convert 60^O to radian measure, multiply 60^O by $\dfrac{\pi}{180}$.

$$60^O \times \frac{\pi}{180^O} = \frac{\pi}{3} \text{ rad}$$

Answer: B

SOLUTION

17. The expression 15 – 3[2 + 6(–3)] simplifies to

(A) -45 (B) -33 (C) 63 (D) 192

Solution:

$$15-3[2+6(-3)]=15-3[2-18]$$

$$=15-3[-16]$$

$$=15+48$$

$$=63$$

Answer: C

18. Ano ang halaga ng $\sum_{m=1}^{3}(2m+1)^{m-1}$?

(A) 15 (B) 55 (C) 57 (D) 245

Solution:

$$\sum_{m=1}^{3}(2m+1)^{m-1}=3^0+5^1+7^2$$

$$=1+5+49$$

$$=55$$

Answer: B

SOLUTION

19. Ang pagsusulit sa asignaturang HEKASI ay may 10 katanungan na nagkakahalaga ng 5 puntos bawat isa, 7 mga katanungan na nagkakahalaga ng 6 na puntos sa bawat isa, at 4 na mga katanungan na nagkakahalaga ng 2 puntos sa bawat isa. Wala sa mga tanong na ito ang bibigyan ng bahagyang kredito. Gaano karaming mga puntos sa pagitan ng 0 at 100 ang imposibleng iskor?

(A) 3 (B) 2 (C) 4 (D) 7

Solution:

- There are 10 questions worth 5 points each so multiples of 5 including 0 up to 50 are possible scores. [5x0=0, 5x1=5, 5x2=10,…,5x10=50].We encircle the numbers.

0	1	2	3	4	5	6	7	8	9	10
	11	12	13	14	15	16	17	18	19	20
	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39	40
	41	42	43	44	45	46	47	48	49	50
	51	52	53	54	55	56	57	58	59	60
	61	62	63	64	65	66	67	68	69	70
	71	72	73	74	75	76	77	78	79	80
	81	82	83	84	85	86	87	88	89	90
	91	92	93	94	95	96	97	98	99	100

- We add multiples of 6 from 6 to 42 to the encircled numbers because there are 7 question worth 6 points each. We underline the resulting numbers.

0	1	2	3	4	5	*6*	7	8	9	10
	11	*12*	13	14	15	*16*	*17*	*18*	19	20
	21	*22*	*23*	*24*	25	*26*	*27*	*28*	*29*	*30*
	31	*32*	*33*	*34*	*35*	*36*	*37*	*38*	*39*	*40*
	41	*42*	*43*	*44*	*45*	*46*	*47*	*48*	*49*	*50*
	51	*52*	*53*	*54*	*55*	*56*	*57*	*58*	*59*	*60*
	61	*62*	*63*	*64*	*65*	*66*	*67*	*68*	*69*	*70*
	71	*72*	73	*74*	*75*	*76*	*77*	78	79	*80*
	81	*82*	83	84	85	*86*	*87*	88	89	90
	91	*92*	93	94	95	96	97	98	99	100

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- We add multiples of 2 from 2 to 8 to the encircled numbers and also to the underlined numbers because there are 4 questions worth 2 points each. We highlight the resulting numbers.

0	1	2	3	4	5	6	7	8	9	10
	11	12	13	14	15	16	17	18	19	20
	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39	40
	41	42	43	44	45	46	47	48	49	50
	51	52	53	54	55	56	57	58	59	60
	61	62	63	64	65	66	67	68	69	70
	71	72	73	74	75	76	77	78	79	80
	81	82	83	84	85	86	87	88	89	90
	91	92	93	94	95	96	97	98	99	100

If the number is not encircled, highlighted or underlined then the number is an impossible score. The impossible scores between 0 and 100 are 1, 3, 97, and 99.

There are four impossible scores between 0 and 100.
Answer: C

SOLUTION

20. Ang isang malaking istante ng libro ay maaaring naglalaman sa pagitan ng 57 at 564 na mga libro. Eksaktong $\frac{1}{6}$ ay librong matematika at eksaktong $\frac{1}{9}$ ay librong physics. Ano ang positibong kaibahan sa pagitan ng pinakamataas at ang pinakamaliit na posibleng bilang ng mga libro na maaaring naka-imbak sa istante?

(A) 468 (B) 486 (C) 504 (D) 522

Solution:

Eksaktong $\frac{1}{6}$ ay librong matematika $\Rightarrow$ the number of books in Mathematics is divisible by 6

eksaktong $\frac{1}{9}$ ay librong physics $\Rightarrow$ the number of books in Physics is divisible by 9

Number of Math and Physics books combined is divisible by both 6 and 9. Therefore, total number of books is divisible by 18.

The smallest number more than 57 that is divisible by 18 is 72.

The largest number less than 564 that is divisible by 18 is 558.

The difference between the largest and smallest possible number of books is 558 – 72 = 486

Answer: B

SOLUTION

21. Simplify: $2\sqrt{\frac{5}{3}} - \sqrt{60} - 5\sqrt{\frac{3}{5}}$

(A) $\frac{-29\sqrt{15}}{15}$ (B) $\frac{-7\sqrt{15}}{3}$ (C) $-\frac{7\sqrt{15}}{15}$ (D) $\frac{-29\sqrt{15}}{3}$

Solution:

First term: $2\sqrt{\frac{5}{3}}$

$= \frac{2\sqrt{5}}{\sqrt{3}}$, Multiply the numerator and denominator by $\sqrt{3}$ to rationalize.

$= \frac{2\sqrt{5}}{\sqrt{3}} \bullet \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{2\sqrt{15}}{3}$

Second Term : $\sqrt{60}$

$= \sqrt{4 \bullet 15}$

$= \sqrt{4} \bullet \sqrt{15}$

$= 2\sqrt{15}$

Third term: $5\sqrt{\frac{3}{5}}$

$= \frac{5\sqrt{3}}{\sqrt{5}} \bullet \frac{\sqrt{5}}{\sqrt{5}}$

$$= \frac{\cancel{5}\sqrt{15}}{\cancel{5}}$$

$$= \sqrt{15}$$

$$2\sqrt{\frac{5}{3}} - \sqrt{60} - 5\sqrt{\frac{3}{5}} = \frac{2\sqrt{15}}{3} - 2\sqrt{15} - \sqrt{15}$$

LCD is 3

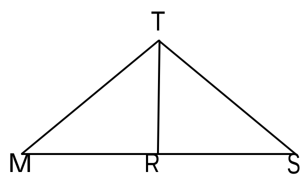
$$= \frac{-7\sqrt{15}}{3}$$

Answer: B

22. Given: R is the midpoint of $\overline{MS}$

$\overline{TR} \perp \overline{MS}$

If you outlined a proof that shows $\overline{TM} \cong \overline{TS}$, which would **NOT** be used?

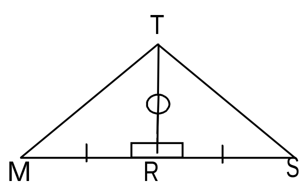


(A) $\Delta TMR \cong \Delta TSR$ by the SAS congruency postulate

(B) $\overline{TM} \cong \overline{TS}$ by CPCTC

(C) $\Delta TMR \cong \Delta TSR$ by the ASA congruency postulate

(D)



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Solution:

Below is a proof that shows $\overline{TM} \cong \overline{TS}$

Statement	Reason	
1. R is the midpoint of $\overline{MS}$	1. Given	S
2. $\overline{MR} \cong \overline{RS}$	2. A midpoint cuts a segment into two congruent segments	
3. $TR \perp MS$	3. Given	A
4. $\angle$MRT and $\angle$SRT are right angles	4. $\perp$ lines form right $\angle$s	
5. $\angle MRT \cong \angle SRT$	5. All right $\angle$s are $\cong$	
6. $\overline{TR} \cong \overline{TR}$	6. reflexive	S
7. $\Delta TMR \cong \Delta TSR$	7. SAS	
8. $\overline{TM} \cong \overline{TS}$	8. CPCTC	

All of the choices were used to prove that

$$\overline{TM} \cong \overline{TS}$$

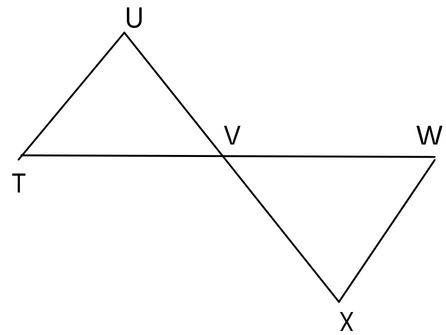
EXCEPT (C) $\Delta TMR \cong \Delta TSR$ by the ASA congruency postulate

ANSWER: C

SOLUTION

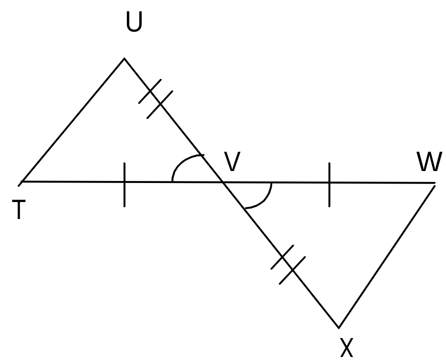
23. Refer to the figure shown. State the congruency postulate that can be used to prove that $\Delta TUV \cong \Delta WXV$.

Given: $\overline{TV} \cong \overline{WV}$ and $\overline{UV} \cong \overline{XV}$



(A) SSS (B) SAS (C) ASA (D) AAS

Solution:

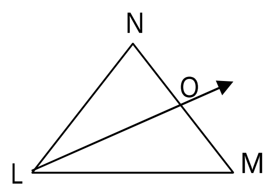


$\Delta TUV \cong \Delta WXV$

Included angle and two included sides are congruent. Therefore by **SAS**

Answer: B **not C as given in the answer key**

24. Find OM if $\overrightarrow{LO}$ bisects $\angle NLM$, $LM = 20$, $NO = 3$, . and $LN = 5$.



(A) 10.23 (B) 0.75 (C) 12 (D) 33.33

Solution:

$$\frac{\text{OM}}{LM} = \frac{NO}{LN}$$

$$\Rightarrow \frac{\text{OM}}{20} = \frac{3}{5}$$

$$\Rightarrow OM = 12$$

ANSWER: C

SOLUTION

25. What value of x will give the maximum value for $-7x^2+7x+3$?

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Solution:

Let $y=-7x^2+7x+3$

$a=-7$, $b=7$, $c=3$

Since a is negative, then

$$h=\frac{-b}{2a}$$

$$=\frac{-7}{-14}$$

$=\frac{1}{2}$, will give the maximum value for $-7x^2+7x+3$.

ANSWER: C

Recall:

In $y=ax^2+bx+c$, the vertex is (h,k)

If a is positive then h will give the minimum value for y , and the minimum value is equal to k .

If a is negative then h will give the maximum value for y , and the maximum value is equal to k .

$$h=\frac{-b}{2a}, \qquad k=\frac{4ac-b^2}{4a}$$

SOLUTION

26. Written in simplest form $\dfrac{x^2y-4}{4-x^2y}$ is

(A) 1 (B) 0 (C) $\dfrac{x^2y-4}{4-x^2y}$ (D) -1

Solution:

$$\frac{x^2y-4}{4-x^2y}=\frac{\cancel{x^2y-4}}{-\left(\cancel{x^2y-4}\right)}$$

$$=-1$$

ANSWER: D

27. Which expression is equivalent to $\dfrac{\sqrt{7}+\sqrt{2}}{\sqrt{7}-\sqrt{2}}$?

(A) $\dfrac{9}{5}$ (B) -1 (C) $\dfrac{9+2\sqrt{14}}{5}$ (D) $\dfrac{11+\sqrt{2}}{14}$

Solution:

Multiply the numerator and denominator by the conjugate of the denominator.

The conjugate of $\sqrt{7}-\sqrt{2}$ is $\sqrt{7}+\sqrt{2}$

$$\frac{\sqrt{7}+\sqrt{2}}{\sqrt{7}-\sqrt{2}}\bullet\frac{\sqrt{7}+\sqrt{2}}{\sqrt{7}+\sqrt{2}}=\frac{7+2\sqrt{14}+2}{7-2}$$

$$=\frac{9+2\sqrt{14}}{5}$$

ANSWER: C

28. Given two lines whose equations are $3x + y - 8 = 0$ and $-2x + ky + 9 = 0$, determine the value of k such that the two lines are perpendicular.

(A) $-\frac{2}{3}$ (B) 6 (C) 8 (D) -9

Solution:

> Recall:
>
> In $y = mx + b$, m is the slope of the line
>
> Two lines are perpendicular if their slopes are negative reciprocals of each other.
>
> or if the product of their slopes is equal to -1.

We express $3x + y - 8 = 0$ in the form $y = mx + b$, the result is $y = -3x + 8$
So its slope is -3.

We express $-2x + ky + 9 = 0$ in the form $y = mx + b$, the result is $y = \frac{2}{k}x - \frac{9}{k}$

So its slope is $\frac{2}{k}$.

We get the product of their slopes and equate to -1.

$(-3)\left(\frac{2}{k}\right) = -1$, if we solve for k, the answer is

$k = 6$

Answer: B

SOLUTION

29. Solve for x: $256^{2x} = 64^{x-2}$

(A) $-\frac{6}{11}$ (B) $-\frac{6}{5}$ (C) $-\frac{1}{5}$ (D) 0

Solution:

Express both sides of the equation in the same base

$$256^{2x} = 64^{x-2}$$

$$\Rightarrow \left(4^4\right)^{2x} = \left(4^3\right)^{x-2}$$

$$\Rightarrow 4^{8x} = 4^{3x-6}$$

$$\Rightarrow 8x = 3x - 6$$

$$\Rightarrow 5x = -6$$

$$\Rightarrow x = \frac{-6}{5}$$

Recall: law of exponent for powers

$\left(a^m\right)^n = a^{mn}$

Answer: B

SOLUTION

30. Find the square root of $x^4 + 2x^3 + 5x^2 + 4x + 4$.

(A) $x^2 + x + 2$ (B) $x^2 + 2x + 2$ (C) $x^2 + 3x + 2$ (D) $x^2 + 2$

Solution:

$$(x^2 + x + 2)^2 = x^4 + 2x^3 + 5x^2 + 4x + 4$$

Answer: A

31. The product of the square roots of two consecutive positive numbers is $2\sqrt{14}$, what is their sum?

(A) 15 (B) 17 (C) 19 (D) 21

Solution:

Let x and $x+1$ be the two consecutive positive numbers.

$$\sqrt{x} \bullet \sqrt{x+1} = 2\sqrt{14}$$

$$\sqrt{x(x+1)} = \sqrt{4}\sqrt{14} = \sqrt{56}$$

$x(x+1) = 56$, so $x = 7$ because 7(7+1) = 56

The two consecutive positive numbers are 7 and 8. Their sum is 15.

Answer: A

SOLUTION

32. Given the formula $C = \frac{5}{9}(F - 32)$; find F when C is 20.

(A) 15 (B) 17 (C) 68 (D) 21

Solution:

$$C = \frac{5}{9}(F - 32)$$

$$20 = \frac{5}{9}(F - 32)$$ multiply both sides by $\frac{9}{5}$

$$\frac{9}{\cancel{5}} \bullet \overset{4}{\cancel{20}} = \cancel{\frac{9}{5}} \bullet \cancel{\frac{5}{9}}(F - 32)$$

$$36 = F - 32$$

$$F = 68$$

Answer: C

33. What number added to 6% of itself equals 31.8?
(A)29.892 (B) 31.74 (C)30 (D) 31

Solution:

Let x be the number

$$x + 0.06x = 31.8$$

$$1.06x = 31.8$$

$$x = \frac{31.8}{1.06}$$

$$x = 30$$

Answer: C

UPCAT REVIEWER

PRACTICE TEST 1

SOLUTION

34. Perform the indicated operations: $(2a-3)^2 - 3a(a-2) - (3-a)^2$

    (A) 2 (B) 0 (C) -3 (D) $2a$

Solution:

$$(2a-3)^2 - 3a(a-2) - (3-a)^2 = 4a^2 - 12a + 9 - 3a^2 + 6a - (a^2 - 6a + 9)$$

$$\begin{array}{r} 4a^2 - 12a + 9 \\ -3a^2 + 6a \\ -a^2 + 6a - 9 \\ \hline 0 \end{array}$$

Answer: B

35. What must be the value of m if $x-5$ is a factor of $2x^2 - mx - 35$?

    (A) 3 (B) 5 (C) 7 (D) 10

Solution:

Let $f(x) = 2x^2 - mx - 35$

$x-5$ is a factor of $f(x) = 2x^2 - mx - 35$ when $f(5) = 0$ [We use the Remainder Theorem]

So $f(5) = 2(5)^2 - 5m - 35$

$= 15 - 5m$

$15 - 5m = 0 \Rightarrow m = 3$

Answer: A

UPCAT REVIEWER

PRACTICE TEST 1

SOLUTION

36. Reduce $\frac{b+a}{b-a} - 2\left(\frac{b}{a} - \frac{b}{a-b}\right)$ to a single fraction in its lowest terms.

(A) $\frac{a+2b}{b}$ (B) $\frac{-2a-b}{a}$ (C) $\frac{-a-2b}{a}$ (D) $\frac{2a+b}{b}$

Solution:

$\frac{b+a}{b-a} - 2\left(\frac{b}{a} - \frac{b}{a-b}\right)$, we distribute -2

$$= \frac{b+a}{b-a} - \frac{2b}{a} + \frac{2b}{a-b}$$

$\frac{b+a}{b-a}$ is also equal to $-\frac{a+b}{a-b}$

$$= -\frac{2b}{a} + \frac{2b}{a-b} - \frac{a+b}{a-b}$$

Similar fractions, [they have the same denominators] so we combine the numerators.

$$= -\frac{2b}{a} + \frac{2b-(a+b)}{a-b}$$

$$= -\frac{2b}{a} + \frac{b-a}{a-b}$$

$$= -\frac{2b}{a} - \frac{\cancel{b-a}}{\cancel{b-a}} \quad 1$$

$$= -\frac{2b}{a} - \frac{a}{a}$$

$$\boxed{1 = \frac{a}{a}}$$

$$= \frac{-a-2b}{a}$$

Answer: C

37. Find the quotient if $2x^3 - 3x^2 - 5x + 6$ is divided by $x^2 - 3x + 2$.

(A) $2x-3$ (B) $2x+3$ (C) $-2x+3$ (D) $-2x-3$

Solution:

$$
\begin{array}{r|l}
 & 2x \quad + 3 \\
\hline
x^2 - 3x + 2 & 2x^3 - 3x^2 - 5x + 6 \\
- & 2x^3 - 6x^2 + 4x \\
\hline
 & \quad\;\; 3x^2 - 9x + 6 \\
 & \quad\;\; 3x^2 - 9x + 6 \\
\hline
 & \qquad\qquad 0
\end{array}
$$

Step 1.Divide $2x^3$ by x^2, the result is $2x$

Step 2.Multiply $2x$ by $x^2 - 3x + 2$, the result is $2x^3 - 6x^2 + 4x$

Step 3.Subtract $2x^3 - 6x^2 + 4x$ from $2x^3 - 3x^2 - 5x + 6$, the result is $3x^2 - 9x$

Step 4.Bring down +6

Step 5.Divide $3x^2$ by x^2, the result is 3

Step 6.Multiply 3 by $x^2 - 3x + 2$, the result is $3x^2 - 9x + 6$

Step 7.Subtract $3x^2 - 9x + 6$ from $3x^2 - 9x + 6$, the result is 0, there is no remainder.

Last Step8. The quotient is $2x + 3$

Answer: B

SOLUTION

38. Ang mga bahay sa Tinio Street ay may sunud-sunod na bilang mula 1 hanggang 447. Ilang tanso na numero ang kailangan upang magawa ang lahat ng bilang ng mga bahay ?

(A) 1232 (B) 1231 (C) 1236 (D) 1233

Solution:

		count
one-digit numbers	1 to 9	9
two-digit numbers	10 to 99	2 x 90 = 180
three-digit numbers	100 to 447	3 x 348 = 1044
	total	**1233**

ANSWER: D

39. Solve for x: $\frac{7x}{5}-\frac{1}{14}(x-11)=\frac{3}{7}(x-25)+34$

(A) 4 (B) 11 (C) 18 (D) 25

Solution:

$$\frac{7x}{5}-\frac{1}{14}(x-11)=\frac{3}{7}(x-25)+34$$

multiply both sides by 70 = LCD

$$(70)\left[\frac{7x}{5}-\frac{1}{14}(x-11)\right]=(70)\left[\frac{3}{7}(x-25)+34\right]$$

distribute 70

$$98x-5(x-11)=30(x-25)+2380$$
$$98x-5x+55=30x-750+2380$$
$$93x+55=30x+1630$$
$$63x=1575$$
$$x=25$$

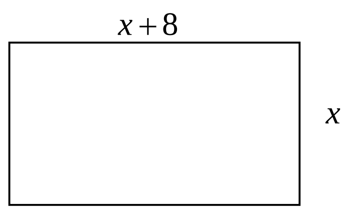
ANSWER: D

SOLUTION

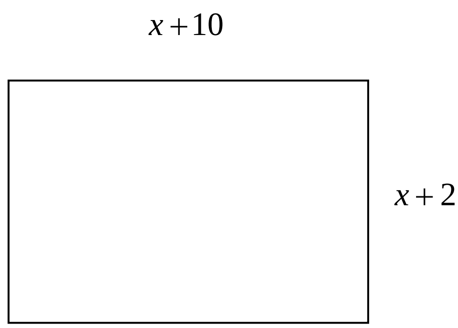
40. The length of a room is 8 feet greater than its width; if each dimension is increased by 2 feet, the area will be increased by 60 square feet. Find the area of the floor.

(A) 65 (B) 105 (C) 153 (D) 180

Solution:



Area = $x(x+8)$



New Area = $(x+2)(x+10) = x(x+8)+60$

$$\Rightarrow (x+2)(x+10) = x(x+8)+60$$

$$\Rightarrow \cancel{x^2} + 12x + 20 = \cancel{x^2} + 8x + 60$$

$$\Rightarrow 4x = 40$$

$$\Rightarrow x = 10$$

Original area of the floor = $10(18) = 180\ ft^2$

ANSWER: D

UPCAT REVIEWER

PRACTICE TEST 1

SOLUTION

41. Find the greatest common factor of $3x^2+6x-9$, $6x^2-21x+15$, and $6x^3-6$.

(A) $3(x-1)$ (B) $3(x+3)$ (C) $3(x+1)$ (D) $3(x-3)$

Solution:

$$3x^2+6x-9=3(x^2+2x-3)=3(x+3)(x-1)$$

$$6x^2-21x+15=3(2x^2-7x+5)=3(2x-5)(x-1)$$

$$6x^3-6=6(x^3-1)=2\cdot 3(x-1)(x^2+x+1)$$

GCF = $3(x-1)$

Answer: A

42. Find the equation of the line that passes through the point (-2,-5) and is parallel to the line 5x-4y=2.

(A) 5x-4y=-17 (B) -2x-5y=2 (C) 5x+4y=2 (D) 5x-4y=10

Solution:

The lines 5x-4y=2 and 5x-4y=k [k is constant] are parallel.

To solve for k, we substitute (-2,-5) in 5x-4y=k

$$5(-2)-4(-5)=k$$

$$-10+20=k$$

$$k=10$$

Therefore the equation of the line that passes through the point (-2,-5) and is parallel to the line 5x-4y=2 is $5x-4y=10$

ANSWER: D

UPCAT REVIEWER

PRACTICE TEST 1

SOLUTION

43. Simplify: $\left(\dfrac{4x^{-3}p^{2}}{y^{-2}}\right)^{-2}\left(\dfrac{y^{3}p^{5}}{x^{2}}\right)^{-3}$

(A) $\dfrac{x^{6}y^{4}}{16p^{14}}$ (B) $\dfrac{-16x^{12}}{y^{11}p^{15}}$ (C) $\dfrac{x^{6}}{4y^{10}p^{19}}$ (D) $\dfrac{x^{12}}{16y^{13}p^{19}}$

Solution:

$$\left(\frac{4x^{-3}p^{2}}{y^{-2}}\right)^{-2}\left(\frac{y^{3}p^{5}}{x^{2}}\right)^{-3}=\left(\frac{4^{-2}x^{6}p^{-4}}{y^{4}}\right)\left(\frac{y^{-9}p^{-15}}{x^{-6}}\right)$$

$$=\left(\frac{4^{-2}x^{6}p^{-4}}{y^{4}}\right)\left(\frac{y^{-9}p^{-15}}{x^{-6}}\right)$$

$$=\frac{x^{12}}{16y^{13}p^{19}}$$

ANSWER: D

44. Simplify: $\left(6x^{1/2}-7y^{7/2}\right)\left(6x^{1/2}+7y^{7/2}\right)$

(A) $36x-42xy+49y^{7}$ (B) $36x-49y^{7}$ (C) $36x+42xy-49y^{7}$ (D) $36x+49y^{7}$

Solution:

$$\left(6x^{1/2}-7y^{7/2}\right)\left(6x^{1/2}+7y^{7/2}\right)=\left(6x^{1/2}\right)^{2}-\left(7y^{7/2}\right)^{2}$$

$$=36x-49y^{7}$$

Recall:

$(a-b)(a+b)=a^{2}-b^{2}$

ANSWER: B

45. Simplify: $\dfrac{\sqrt[3]{x^6y^3}\,xy^4}{\sqrt{x^2y^8}\,(xy)^{-3}}$

(A) x^4y^3 (B) xy (C) x^5y^6 (D) x^5y^4

Solution:

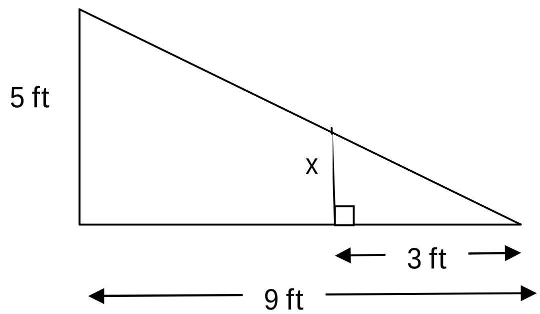
$$\frac{\sqrt[3]{x^6y^3}\,xy^4}{\sqrt{x^2y^8}\,(xy)^{-3}} = \frac{(x^2y)\cancel{(xy^4)}(x^3y^3)}{\cancel{xy^4}}$$

$$= x^5y^4$$

ANSWER: D

SOLUTION

46. Use similar triangles to find x.



(A) 8/9 ft (B) 5.4 ft (C) 15 ft (D) $1\frac{2}{3}$ *ft*

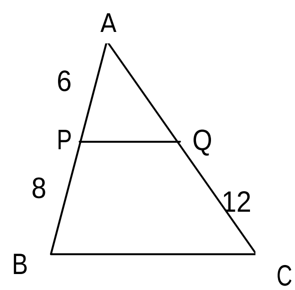
Solution:

$$\frac{x}{3}=\frac{5}{9}$$

$$x=\frac{5}{3} \text{ or } 1\tfrac{2}{3} \text{ ft}$$

ANSWER: D

47. Given: $\overline{PQ} \,//\, \overline{BC}$. Find the length of $\overline{AC}$.



(A) 17 (B) 21 (C) 23 (D) 18

Solution:

$$\frac{\text{AP}}{\text{AB}} = \frac{AQ}{AC}$$ Let AQ = x

$$\frac{6}{14} = \frac{x}{x+12}$$ cross multiply

$$14x = 6x + 72$$

$$8x = 72$$

$$x = 9$$

AC = x+12 = 9 + 12 = 21

Answer: B

SOLUTION

48. The numbers 27, 36, and 45 represents the length of the sides of a/an

(A) acute triangle (B) obtuse triangle (C) no triangle (D) right triangle

Solution:

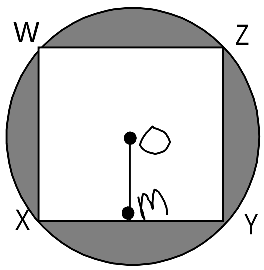
3-4-5 is a Pythagorean Triple, multiply by 9

27-36-45

Therefore the numbers 27, 36, and 45 represents the length of the sides of a right triangle.

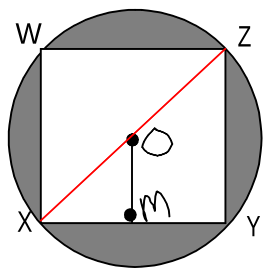
ANSWER: D

49. In the figure shown, square WXYZ is inscribed in circle O. Also, $\overline{OM} \perp \overline{XY}$ and $OM = 7$. Find the area of the shaded region.



(A) $49\pi - 49$ (B) $49\sqrt{2}\pi - 49$ (C) $98\pi - 196$ (D) $147\pi - 196$

Solution:



$YZ = 2 \times OM = 14$

XZ is a diagonal of the square, therefore

$XZ = \sqrt{2} \times YZ = 14\sqrt{2}$

XZ is also the diameter of the circle, so the radius is one-half of its measure.

Radius of circle O = $7\sqrt{2}$

Area of shaded region = Area of circle – Area of square

$$= \pi\left(7\sqrt{2}\right)^2 - 14^2$$

$$= 98\pi - 196$$

Answer: C

SOLUTION

50. Simplify: $\dfrac{\frac{5}{6}+\frac{1}{3}}{2-\left(\frac{7}{8}-\frac{1}{3}\right)}$

(A) $\dfrac{3}{8}$ (B) $\dfrac{4}{5}$ (C) $\dfrac{2}{3}$ (D) $\dfrac{1}{6}$

Solution:

$$\frac{\frac{5}{6}+\frac{1}{3}}{2-\left(\frac{7}{8}-\frac{1}{3}\right)}\bullet\frac{24}{24}=\frac{20+8}{48-24\left(\frac{7}{8}-\frac{1}{3}\right)}$$

$$=\frac{28}{48-21+8}$$

$$=\frac{28}{35}$$

$$=\frac{4}{5}$$

Answer: B

SOLUTION

51. Evaluate $x^2 - 2y^2 + 4(x-y)(2x^5 + 4xy^5 + 3y^5)^0$ if $x = 3$ and $y = 4$.

(A) -45 (B) -81 (C) -36 (D) -27

Solution:

$$x^2 - 2y^2 + 4(x-y)\underbrace{(2x^5 + 4xy^5 + 3y^5)^0}_{1}$$

$= x^2 - 2y^2 + 4(x-y)$ substitute $x = 3$ and $y = 4$.

$= 3^2 - 2(4)^2 + 4(3-4)$

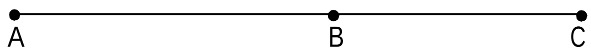
$= 9 - 32 + 4(-1)$

$= -27$

ANSWER: D

SOLUTION

52. The length of $\overline{AC}$ is $5\frac{1}{6}$ meters. The length of $\overline{BC}$ is $2\frac{1}{2}$ meters. Find AB.



(A) 2 m (B) $7\frac{1}{4}$ m (C) $7\frac{2}{3}$ m (D) $2\frac{2}{3}$ m

Solution:

$$AB + BC = AC$$

$$AB + 2\frac{1}{2} = 5\frac{1}{6}$$

$$AB = 5\frac{1}{6} - 2\frac{1}{2}$$

$$= 5\frac{1}{6} - 2\frac{3}{6}$$

$$= \frac{31}{6} - \frac{15}{6}$$

$$= \frac{16}{6}$$

$$= \frac{8}{3} \text{ or } 2\frac{2}{3} \text{ m}$$

ANSWER: D

UPCAT REVIEWER

PRACTICE TEST 1

SOLUTION

53. What is the sum of $\sqrt{-2}$ and $\sqrt{-18}$?

(A) $2i\sqrt{5}$ (B) $5i\sqrt{2}$ (C) $4i\sqrt{2}$ (D) $6i$

Solution:

$$\sqrt{-2}+\sqrt{-18}=i\sqrt{2}+3i\sqrt{2}$$

$$=4i\sqrt{2}$$

Answer: C

54. Ano ang ika-7-term sa isang geometric sequence kung ang unang term ay 81 at ang ika-11-term ay $\frac{1}{729}$?

(A) $\frac{1}{27}$ (B) $\frac{1}{9}$ (C) $\frac{1}{3}$ (D) 1

Solution:

Given: $a_1 = 81,\ a_{11} = \frac{1}{729},\ a_7 = ?$

In geometric sequence the nth term is $a_n = a_1 r^{n-1}$

$$a_{11} = a_1 r^{11-1}$$

$$\frac{1}{729} = 81r^{10}$$

$$\frac{1}{729 \cdot 81} = r^{10}$$

$$\frac{1}{3^6 \cdot 3^4} = r^{10}$$

$$\frac{1}{3^{10}} = r^{10}$$

$$\left(\frac{1}{3}\right)^{10} = r^{10}$$

$$r = \frac{1}{3}$$

Therefore

$$a_7 = a_1 r^{7-1}$$

$$= (81)\left(\frac{1}{3}\right)^6$$

$$= \frac{3^4}{3^6}$$

$$= \frac{1}{3^2}$$

$$= \frac{1}{9}$$

Answer: B

55. Kung ang 25% ng isang numero ay 75. Ano naman ang 30% ng numero?

1

(A) 80 (B) 90 (C) 100 (D) 85

Solution:

Let x be the number

$0.25x = 75$, multiply both sides by 4

$x = 300$

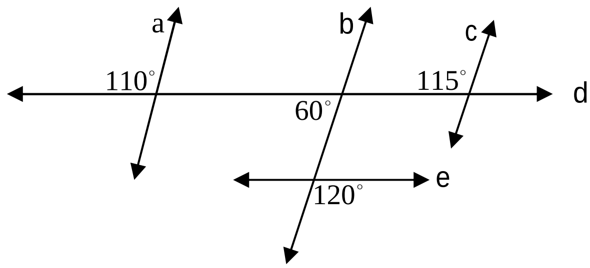
Therefore

Answer: B

$0.3x = 90$

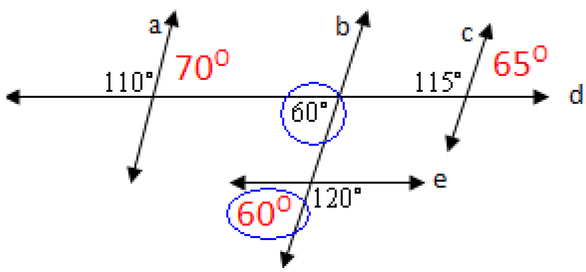
SOLUTION

56. Based on the diagram below, which statement is true?



(A) a // b (B) a // c (C) b // c (D) d // e

Solution:



Therefore d//e

ANSWER: D

SOLUTION

57. Ayon sa isang sinaunang paniniwala, kapag ang isang kaibigan ay dumalaw sa isang may sakit na tao, $1/30$ ng kanyang pagkakasakit ay nawawala . Ano ang pinakamababang bilang ng kaibigan ang kailangang bumisita sa may sakit upang maalis ang 98% o higit pa ng kanyang pagkakasakit?

(A) 114 (B) 115 (C) 116 (D) 117

Solution:

$$\boxed{1-\frac{1}{30}}$$

$$\boxed{100\% - 98\%}$$

$$\left(\frac{29}{30}\right)^x = 0.02$$

get the log of both sides

$$x \cdot \log\left(\frac{29}{30}\right) = \log 0.02$$

Recall:

$$\log_b a^n = n \cdot \log_b a$$

$$x = \frac{\log 0.02}{\log\left(\frac{29}{30}\right)}$$

$$x = 115.3936$$

The smallest integer greater than 115.3936 is 116.

Answer: C

58. Si Sarah ay gagawa ng isang keyk at ilang mga cookies. Ang keyk ay nangangailangan ng $3/8$ tasa ng asukal at ang mga cookies ay nangangailangan ng $3/5$ tasa ng asukal. Si Sarah ay may $15/16$ tasa ng asukal. Siya ba ay may sapat na asukal?

(A) Siya ay may sapat na asukal
(B) Kailangan pa niya ng $1/8$ tasa ng asukal.
(C) Kailangan pa niya ng $3/80$ tasa ng asukal.
(D) Kailangan pa niya ng $4/19$ tasa ng asukal.

Solution:

Sarah needs $\left(\frac{3}{8}+\frac{3}{5}\right)-\frac{15}{16}$ cups of sugar

$$=\frac{39}{40}-\frac{15}{16}$$

$$=\frac{39}{40}-\frac{15}{16}$$

$$=\frac{78-75}{80}$$

$$=\frac{3}{80}$$

Answer: C

SOLUTION

59. Find the distance from the point (2,3) to the line $x - y = 5$.

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $3\sqrt{2}$

Solution:

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|(1)(2) + (-1)(3) + (-5)|}{\sqrt{1^2 + (-1)^2}}$$

$$= \frac{6}{\sqrt{2}}$$

$$= 3\sqrt{2}$$

ANSWER: D

SOLUTION

60. If $x+\frac{1}{x}=4$, what is the value of $x^2+\frac{1}{x^2}$?

(A) 16 (B) 15 (C) 14 (D) 12

Solution:

$x+\frac{1}{x}=4$ get the square of both sides

$$x^2+2\cdot\cancel{x}\cdot\frac{1}{\cancel{x}}+\frac{1}{x^2}=16$$

$$x^2+\frac{1}{x^2}=14$$

Answer: C

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